Rossmoyne Senior High School

Semester One Examination, 2023

Question/Answer booklet

MATHEMATICS APPLICATIONS UNIT 1

Section One: Calculator-free

SOLUTIONS

WA student number: In figures

In words

Circle your Teacher's Name: Mr Fletcher Mrs Fraser-Jones Mrs Murray Mr Pisano Mr Stephens

Time allowed for this section

Reading time before commencing work: five minutes
Working time: fifty minutes

Materials required/recommended for this section

To be provided by the supervisor
This Question/Answer booklet
Formula sheet

To be provided by the candidate
Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: nil

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	7	7	50	52	35
Section Two: Calculator-assumed	12	12	100	98	65
				Total	100

Instructions to candidates

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.

2. Write your answers in this Question/Answer booklet preferably using a blue/black pen. Do not use erasable or gel pens.

3. You must be careful to confine your answers to the specific question asked and to follow any instructions that are specific to a particular question.

4. Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.

5. It is recommended that you do not use pencil, except in diagrams.

6. Supplementary pages for planning/continuing your answers to questions are provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.

7. The Formula sheet is not to be handed in with your Question/Answer booklet.

Section One: Calculator-free **35% (52 Marks)**

This section has **seven** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 50 minutes.

Question 1 **(7 marks)**

Consider the matrices $\boldsymbol{P} = \begin{bmatrix} 2 & 0 \\ -1 & 1 \end{bmatrix}$, $\boldsymbol{Q} = \begin{bmatrix} 3 & 0 \end{bmatrix}$, $\boldsymbol{R} = \begin{bmatrix} -3 \\ 0 \end{bmatrix}$ and $\boldsymbol{S} = \begin{bmatrix} 4 & 2 & 0 \\ 1 & 0 & -2 \end{bmatrix}$.

(a) The size of the matrix that has a zero in its second row and first column is $a \times b$. State the value of a and the value of b. (1 mark)

Solution
The size of matrix $\boldsymbol{R}$ is 2×1 and so $a = 2$ and $b = 1$.
Specific behaviours
✓ correct values

(b) Calculate $\boldsymbol{P} + 4\boldsymbol{I}$, where $\boldsymbol{I}$ is the 2×2 identity matrix. (2 marks)

Solution
$\boldsymbol{P} + 4\boldsymbol{I} = \begin{bmatrix} 2 & 0 \\ -1 & 1 \end{bmatrix} + 4\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 6 & 0 \\ -1 & 5 \end{bmatrix}$
Specific behaviours
✓ correctly indicates $4\boldsymbol{I}$
✓ correct matrix result

(c) The sum of matrices $\boldsymbol{S}$ and $\boldsymbol{T}$ is $\begin{bmatrix} 3 & 2 & 1 \\ 3 & 2 & 1 \end{bmatrix}$. Determine matrix $\boldsymbol{T}$. (2 marks)

Solution
$\boldsymbol{T} = \begin{bmatrix} 3 & 2 & 1 \\ 3 & 2 & 1 \end{bmatrix} - \begin{bmatrix} 4 & 2 & 0 \\ 1 & 0 & -2 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 1 \\ 2 & 2 & 3 \end{bmatrix}$
Specific behaviours
✓ at least three correct entries
✓ correct matrix $\boldsymbol{T}$

(d) Calculate the matrix product $\boldsymbol{PS}$. (2 marks)

Solution
$\boldsymbol{PS} = \begin{bmatrix} 2 & 0 \\ -1 & 1 \end{bmatrix} \times \begin{bmatrix} 4 & 2 & 0 \\ 1 & 0 & -2 \end{bmatrix} = \begin{bmatrix} 8 & 4 & 0 \\ -3 & -2 & -2 \end{bmatrix}$
Specific behaviours
✓ at least three correct entries
✓ correct product

Question 2 **(6 marks)**

(a) Given $\frac{ab}{c} + d = e$,

(i) find d if $a = -1, b = 4, c = 1$ and $e = -10$. (2 marks)

Solution
$\frac{-1 \times 4}{1} + d = -10$ $d = -6$
Specific behaviours
✓ correctly substitutes ✓ correct value

(ii) find c if $a = -2, b = 3, d = -4$ and $e = -2$. (2 marks)

Solution
$\frac{-2 \times 3}{c} - 4 = -2$ $c = -3$
Specific behaviours
✓ correctly substitutes ✓ correct value

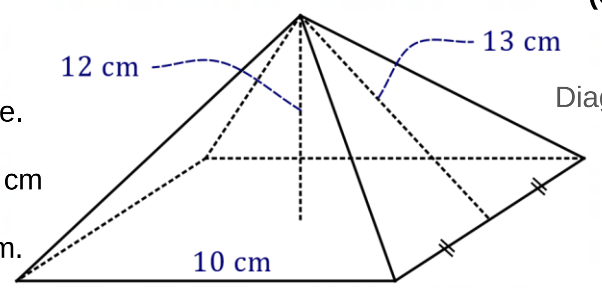
(b) Simplify the expression $\frac{14 + 2x}{x^2 + x - 8}$ when $x = -5$. (2 marks)

Solution
$\frac{14 + 2x}{x^2 + x - 8} = \frac{14 + 2(-5)}{(-5)^2 + (-5) - 8} = \frac{14 - 10}{25 - 5 - 8} = \frac{4}{12} = \frac{1}{3}$
Specific behaviours
✓ correctly evaluates numerator ✓ correctly evaluates denominator -1 mark if not simplified

Question 3 **(9 marks)**

The diagram at right, shows a scale of a right pyramid with its apex 12 cm directly above the centre of its square base.

The length of one side of the square is 10 cm and the distance from the apex to the midpoint of one side of the square is 13 cm.



(a) Clearly label the diagram with the three given dimensions. (1 mark)

Solution
See diagram
Specific behaviours
✓ correctly labels dimensions on diagram

(b) Determine the volume of the pyramid. (2 marks)

Solution
$A_{BASE} = 10 \times 10 = 100$ $V = \frac{1}{3} \times 100 \times 12 = 400 \text{ cm}^3$
Specific behaviours
✓ indicates correct area of base (can be in within 1 step) ✓ correct volume

(c) Determine the total surface area of the pyramid. (3 marks)

Solution
$A_\Delta = \frac{1}{2} \times 10 \times 13$ $4 \times A_\Delta = 4 \times \frac{1}{2} \times 10 \times 13 = 10 \times 26 = 260$ $TSA = A_{4\Delta} + A_{BASE} = 260 + 100 = 360 \text{ cm}^2$
Specific behaviours
✓ indicates correct calculation for area of one triangle ✓ indicates TSA is sum of 4 triangles and base ✓ correct total surface area

Note:
$Award\ first\ 2\ marks\ for$ $2 \times 10 \times 13 = 260 \text{ cm}^2$

(d) A playground pyramid is to be constructed at the local park, based on this model pyramid. The scale ratio of lengths of the model pyramid compared to the playground pyramid is 1:50. Determine the number of square metres of land that the playground pyramid will cover. (3 marks)

Solution
Length of side in playground is $10 \times 50 \div 100 = 5$ m. Hence area covered is $5 \times 5 = 25$ square metres.
Specific behaviours
✓ indicates correct use of scale ✓ converts units ✓ correct number of square metres

Alternative Solution
$1:50$ $10:x$ $\therefore x = 500\ cm = 5m$ Hence area covered is $5 \times 5 = 25$ square metres.
Specific behaviours
✓ indicates correct use of scale ✓ converts units ✓ correct number of square metres

Question 4 **(9 marks)**

The table below shows the buy and sell rates for one Australian dollar advertised by an airport foreign exchange kiosk for several currencies.

Currency (country)	Buy rate	Sell rate
Pound (United Kingdom)	0.6	0.55
Kroner (Swedish)	7.00	6.50
Franc (New Caledonia)	80	72

(a) Explain why the kiosk would use the sell rate if a person wanted to exchange $\$200$ for francs and state how many francs the person would receive. (2 marks)

Solution
Because the kiosk is selling the francs to the person. The person would receive $200 \times 72 = 14\,400$ francs.
Specific behaviours
✓ indicates perspective of kiosk as seller of currency ✓ calculates number of francs

(b) A traveller returned from the United Kingdom with 60 pounds. Determine how many dollars they would receive in exchange from the kiosk. (2 mark)

Solution
Number of dollars is $60 \div 0.6 = \$100$.
Specific behaviours
✓ divides no f/t if they multiplied ✓ correct amount

(c) The kiosk sets the sell rate for the New Caledonian francs at 90% of the value of the buy rate.
If the current buy rate decreased by 5 francs, determine the new sell rate. (2 marks)

Solution
Decrease in sell rate is $5 \times 0.9 = 4.5$ and so new sell rate is $72 - 4.50 = 67.50$.
Specific behaviours
✓ shows correct method ✓ correct sell rate

Alternative Solution
$75 \times 0.1 = \$7.50$ and so new sell rate is $72 - 7.50 = 67.50$.
Specific behaviours
✓finds 10% of new buy rate ✓ correct sell rate

(d) A student exchanged $\$400$ into Swedish kroner at the kiosk. They spent 2320 kroner during their trip to Sweden and exchanged the remaining kroner into dollars at the kiosk on their return to Australia. Determine how many dollars they received. (3 marks)

Solution
Number of kroner sold to student is $400 \times 6.50 = 2600$ Student returns with $2600 - 2320 = 280$ kroner. Kiosk buys back 280 kroner for $280 \div 7 = \$40$.
Specific behaviours
✓ shows calculation for kiosk selling kroner to student ✓ shows calculation for kiosk buying kroner from student ✓ correct number of dollars

Question 5 **(8 marks)**

Dylan has created the spreadsheet shown to model his weekly spending budget.

	A	B
1	Rent	\$180
2	Electricity	
3	Transport	\$55
4	Food	\$210
5	Other	\$130
6	Total	\$600

(a) Determine the amount that Dylan has budgeted for electricity. (2 mark)

Solution
Budgeted amounts shown: $180 + 55 + 210 + 130 = \$575$. Hence, his budget for electricity must be $\$25$.
Specific behaviours
✓ demonstrates correct method. ✓ correct amount

Dylan currently works 4 days per week at a vineyard where he is paid $\$6.50$ cash-in-hand for each box of grapes that he picks.

(b) If Dylan picks 30 boxes of grapes per day, determine how much of his earnings remain each week after he has spent all his budget and put aside 10% of his earnings for tax. (3 marks)

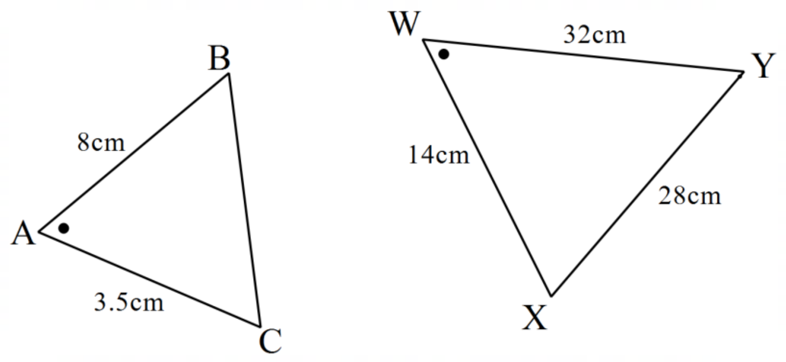
Solution
Weekly earnings: $30 \times 6.50 \times 4 = 30 \times 26 = \$780$. Tax: $780 \times 10\% = \$78$. Hence, he can save $780 - 600 - 78 = \$102$ per week.
Specific behaviours
✓ calculates weekly earnings ✓ calculates amount to set aside for tax ✓ calculates weekly amount saved

(c) Dylan has been told that his rent is going to increase by 15% but plans to reduce his 'other' spending of $\$130$ by 20%. Determine the overall effect that these two changes will have on his weekly spending budget. (3 marks)

Solution
Rent increase: $180 \times 15\% = 18 + 9 = \$27$. Other decrease: $130 \times 20\% = \$26$. After the two changes his weekly spending will increase by $\$1$.
Specific behaviours
✓ calculates rent increase ✓ calculates other decrease ✓ correctly describes weekly spending change

Question 6 **(6 marks)**

These two triangles are similar. Note that they are not drawn to scale.



(a) Write a similarity statement for the triangles above. (2 marks)

Solution
$\Delta ABC \sim \Delta WYX \quad (SAS)$
Specific behaviours
✓ letters in correct order with correct similarity symbol ✓ states similarity test

(b) Determine the length of the missing side. (1 mark)

Solution
$BC = 28 \div 4 = 7\ cm$
Specific behaviours
✓ correct length

(c) The area of ΔWXY is 192cm². Use the length scale factor to calculate the area of ΔABC. (2 marks)

Solution
$ASF = LSF^2 = 4^2 = 16$ $A = 192 \div 16 = 12\ cm^2$
Specific behaviours
✓ correct area scale factor ✓ correct area

Alternate Solution
$SF = 1^2 : 4^2 = 1 : 16$ $A = 192 \div 16 = 12\ cm^2$
Specific behaviours
✓ correct ratio for scale factor ✓ correct area

(d) These triangles are the front view of two similar triangular prisms.
What is the ratio of their volumes? (1 mark)

Solution
$VSF = LSF^3 = 4^3 = 64$
Specific behaviours
✓ correct volume scale factor No mark if left it as 4^3

Alternate Solution
$SF = 1^3 : 4^3 = 1 : 64$
Specific behaviours
✓ correct ratio for scale factor No mark if left it as $1^3 : 4^3$

Question 7 **(7 marks)**

(a) The gear ratio R of a mechanical system is calculated using $R = \frac{d_2}{d_1 + 8} - \frac{d_1}{d_2 - 1}$, where d_1 and d_2 are the diameters of two cogs.

Determine the value of R when $d_1 = 4$ cm and $d_2 = 9$ cm. (2 marks)

Solution
$R = \frac{9}{4+8} - \frac{4}{9-1} = \frac{9}{12} - \frac{4}{8} = \frac{3}{4} - \frac{1}{2} = \frac{1}{4}\ (= 0.25)$
Specific behaviours
✓ substitutes and simplifies denominators ✓ correct simplified value Allow 1 mark for $\frac{9}{12} - \frac{4}{8}$

(b) When an object is placed at a distance of x cm from a lens with focal length of f cm, its image will appear at a distance d cm from the lens, where $d = \frac{xf}{x - f}$. An object is 12.5 cm from a lens that has a focal length of 10 cm. Determine how far its image will appear from the lens. (2 marks)

Solution
$d = \frac{12.5 \times 10}{12.5 - 10} = \frac{125}{2.5} = \frac{250}{5} = 50 \text{ cm}$
Specific behaviours
✓ substitutes and evaluates both parts of fraction ✓ correct distance

(c) The velocity of a model car is given by $v = \sqrt{u^2 + 2as}$ cm/s. Determine

(i) the velocity of the model car when $u = 4, a = 1.5$ and $s = 3$. (1 mark)

Solution
$v = \sqrt{4^2 + 2 \times 1.5 \times 3} = \sqrt{25} = 5 \text{ cm/s}$
Specific behaviours
✓ correct velocity

(ii) the change in velocity of the model car when $u = 4$ and $a = 1.5$, as the value of s increases from 3 to 16. (2 marks)

Solution
$v = \sqrt{4^2 + 2 \times 1.5 \times 16} = \sqrt{64} = 8$ Increase in velocity is $8 - 5 = 3$ cm/s.
Specific behaviours
✓ calculates new velocity ✓ calculates increase

End of questions

Supplementary page

Question number: _________

Supplementary page

Question number: _________

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